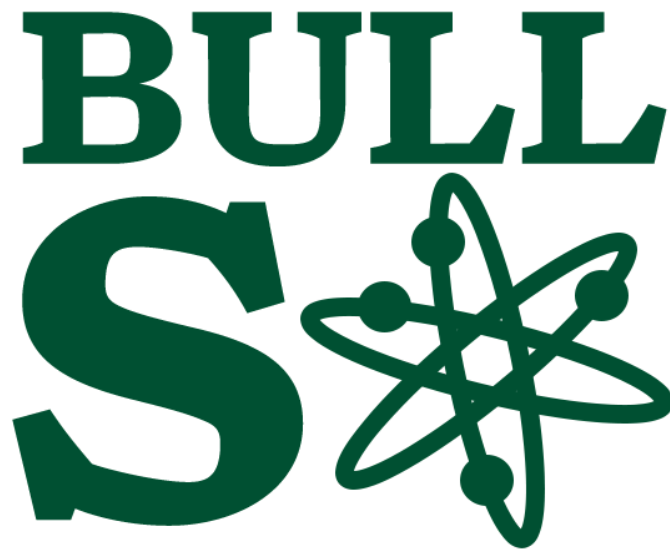


BullSO 2026 Invitational

Heredity B



Bull Science Olympiad at the University of South Florida

Saturday, January 24th, 2026

Team Number: _____

Team Name: _____

Participant Names: _____

Total Score: _____/100

Rank: _____

- | | | |
|-----------|-----------|-----------|
| 1. _____ | 18. _____ | 35. _____ |
| 2. _____ | 19. _____ | 36. _____ |
| 3. _____ | 20. _____ | 37. _____ |
| 4. _____ | 21. _____ | 38. _____ |
| 5. _____ | 22. _____ | 39. _____ |
| 6. _____ | 23. _____ | 40. _____ |
| 7. _____ | 24. _____ | 41. _____ |
| 8. _____ | 25. _____ | 42. _____ |
| 9. _____ | 26. _____ | 43. _____ |
| 10. _____ | 27. _____ | 44. _____ |
| 11. _____ | 28. _____ | 45. _____ |
| 12. _____ | 29. _____ | 46. _____ |
| 13. _____ | 30. _____ | 47. _____ |
| 14. _____ | 31. _____ | 48. _____ |
| 15. _____ | 32. _____ | 49. _____ |
| 16. _____ | 33. _____ | 50. _____ |
| 17. _____ | 34. _____ | |

Section 1: Mendelian Genetics – Laws & Patterns

1. Which statement best describes Mendel's Law of Segregation?
 - A. Alleles assort independently during meiosis
 - B. Each organism carries two alleles for a gene that separate during gamete formation
 - C. Dominant alleles always eliminate recessive alleles
 - D. Genes on the same chromosome are inherited together
 2. A trait that is expressed only when two recessive alleles are present is considered:
 - A. Codominant
 - B. Incompletely dominant
 - C. Recessive
 - D. Sex-linked
 3. In a monohybrid cross between two heterozygous parents ($Aa \times Aa$), what fraction of offspring are expected to be heterozygous?
 - A. $1/4$
 - B. $1/2$
 - C. $3/4$
 - D. All
 4. Which inheritance pattern results in a phenotype that is intermediate between two homozygous parents?
 - A. Codominance
 - B. Complete dominance
 - C. Incomplete dominance
 - D. Epistasis
 5. Which situation violates Mendel's Law of Independent Assortment?
 - A. Genes on different chromosomes
 - B. Linked genes on the same chromosome
 - C. Autosomal dominant traits
 - D. Recessive traits
-

Section 2: Probability & Punnett Squares

6. What is the probability of producing a homozygous recessive offspring from a heterozygous cross ($Aa \times Aa$)?
 - A. 0%
 - B. 25%
 - C. 50%
 - D. 75%

7. In a dihybrid cross ($AaBb \times AaBb$), how many different gamete types can each parent produce?
 - A. 2
 - B. 3
 - C. 4
 - D. 8
 8. If the probability of trait A is $1/2$ and trait B is $1/4$, what is the probability of both occurring together?
 - A. $1/8$
 - B. $1/6$
 - C. $1/4$
 - D. $3/4$
 9. Which rule of probability is used when calculating the chance of two independent events occurring together?
 - A. Addition rule
 - B. Subtraction rule
 - C. Multiplication rule
 - D. Division rule
 10. In a dihybrid cross showing independent assortment, what is the expected phenotypic ratio?
 - A. 3:1
 - B. 1:2:1
 - C. 9:3:3:1
 - D. 1:1:1:1
-

Section 3: Pedigrees & Inheritance Modes

11. A trait that appears in every generation of a pedigree is most likely:
 - A. Autosomal recessive
 - B. Autosomal dominant
 - C. X-linked recessive
 - D. Mitochondrial
12. Which pedigree feature suggests an X-linked recessive trait?
 - A. Equal numbers of affected males and females
 - B. Only females affected
 - C. More males affected than females
 - D. Trait skipping all generations
13. If two unaffected parents have an affected child, the trait is most likely:
 - A. Autosomal dominant

- B. Autosomal recessive
 - C. X-linked dominant
 - D. Codominant
14. Which individual in a pedigree is considered a carrier?
- A. Affected homozygous dominant
 - B. Affected homozygous recessive
 - C. Unaffected heterozygous
 - D. Affected heterozygous
15. Which inheritance pattern is most commonly associated with color blindness?
- A. Autosomal dominant
 - B. Autosomal recessive
 - C. X-linked recessive
 - D. Y-linked
-

Section 4: Mitosis & Meiosis

16. During which phase of mitosis do sister chromatids separate?
- A. Prophase
 - B. Metaphase
 - C. Anaphase
 - D. Telophase
17. Which process produces genetically identical daughter cells?
- A. Meiosis I
 - B. Meiosis II
 - C. Mitosis
 - D. Fertilization
18. What is the chromosome number of human gametes?
- A. 23
 - B. 46
 - C. 92
 - D. Variable
19. Crossing over occurs during which stage of meiosis?
- A. Prophase I
 - B. Metaphase I
 - C. Anaphase II
 - D. Telophase II
20. The primary purpose of meiosis is to:
- A. Repair damaged DNA
 - B. Produce diploid cells

- C. Increase genetic variation
 - D. Grow tissues
-

Section 5: DNA Structure & Replication

- 21. DNA strands are held together primarily by:
 - A. Covalent bonds
 - B. Hydrogen bonds
 - C. Ionic bonds
 - D. Peptide bonds
 - 22. According to Chargaff's rules, the percentage of adenine equals the percentage of:
 - A. Cytosine
 - B. Guanine
 - C. Thymine
 - D. Uracil
 - 23. DNA polymerase adds nucleotides in which direction?
 - A. 3' to 5'
 - B. 5' to 3'
 - C. Both directions
 - D. Randomly
 - 24. Which enzyme unwinds the DNA double helix during replication?
 - A. Ligase
 - B. Primase
 - C. Helicase
 - D. Polymerase
 - 25. The leading strand is synthesized:
 - A. Discontinuously
 - B. In short fragments
 - C. Continuously
 - D. Only after replication ends
-

Section 6: DNA Organization & Mutations

- 26. DNA wrapped around histone proteins forms:
 - A. Plasmids
 - B. Nucleosomes

- C. Ribosomes
 - D. Operons
27. Euchromatin is best described as DNA that is:
- A. Highly condensed and inactive
 - B. Uncondensed and transcriptionally active
 - C. Found only in bacteria
 - D. Circular
28. Which mutation changes one amino acid into another?
- A. Silent
 - B. Nonsense
 - C. Missense
 - D. Frameshift
29. Which mutation is most likely to have the greatest effect on protein structure?
- A. Silent
 - B. Missense
 - C. Frameshift
 - D. Substitution
30. A deletion of one nucleotide in a coding sequence causes:
- A. Codominance
 - B. A frameshift mutation
 - C. A silent mutation
 - D. Incomplete dominance
-

Section 7: Central Dogma

31. The correct flow of genetic information is:
- A. Protein → RNA → DNA
 - B. DNA → RNA → Protein
 - C. RNA → DNA → Protein
 - D. DNA → Protein → RNA
32. Transcription occurs in the:
- A. Cytoplasm
 - B. Ribosome
 - C. Nucleus
 - D. Golgi apparatus
33. Which molecule carries amino acids to the ribosome?
- A. mRNA
 - B. rRNA

- C. tRNA
 - D. DNA
34. Codons are found on:
- A. DNA
 - B. tRNA
 - C. mRNA
 - D. rRNA
35. The start codon for translation is:
- A. UAA
 - B. AUG
 - C. UGA
 - D. UAG
-

Section 8: Transcription & Translation

36. Which enzyme synthesizes RNA during transcription?
- A. DNA polymerase
 - B. RNA polymerase
 - C. Ligase
 - D. Helicase
37. Translation occurs at the:
- A. Nucleus
 - B. Ribosome
 - C. Mitochondrion
 - D. Cell membrane
38. The role of rRNA is to:
- A. Carry genetic instructions
 - B. Form the structure of ribosomes
 - C. Carry amino acids
 - D. Regulate transcription
39. A stop codon signals:
- A. Start of transcription
 - B. End of transcription
 - C. Start of translation
 - D. End of translation
40. Which process requires ATP directly?
- A. DNA replication only
 - B. Transcription only

- C. Translation only
 - D. All of the above
-

Section 9: Regulation & Operons

- 41. The lac operon is an example of:
 - A. Constitutive gene expression
 - B. A repressible operon
 - C. An inducible operon
 - D. A mutation
 - 42. The lac operon is activated when:
 - A. Glucose is abundant
 - B. Lactose is absent
 - C. Lactose is present
 - D. DNA is replicated
 - 43. A repressor protein functions by:
 - A. Binding RNA polymerase
 - B. Blocking transcription
 - C. Destroying mRNA
 - D. Creating mutations
 - 44. Gene regulation allows cells to:
 - A. Express all genes at all times
 - B. Conserve energy and resources
 - C. Increase mutation rates
 - D. Eliminate DNA damage
 - 45. Operons are most commonly found in:
 - A. Eukaryotes
 - B. Prokaryotes
 - C. Viruses
 - D. Mitochondria
-

Section 10: Technology & Techniques

- 46. PCR is primarily used to:
 - A. Sequence proteins
 - B. Amplify DNA

- C. Translate RNA
 - D. Edit genes
47. During PCR, DNA strands separate during the:
- A. Annealing step
 - B. Extension step
 - C. Denaturation step
 - D. Translation step
48. Which component is required for PCR?
- A. Ribosomes
 - B. RNA polymerase
 - C. DNA polymerase
 - D. Restriction enzymes
49. Which technique allows scientists to compare DNA fragment sizes to identify individuals or genetic differences?
- A. PCR
 - B. Gel electrophoresis
 - C. Translation
 - D. Transcription
50. PCR is best suited for answering which question?
- A. How many chromosomes are in a cell?
 - B. Is a specific DNA sequence present in a sample?
 - C. What proteins are expressed?
 - D. How does translation begin?